

## chapter 7 - aluminum scraper

| Aluminum | Temper | Elastic Mod | Shear Mod | Yield Stre | Tensile Strength- $\sigma_u$ -(103psi) |
|----------|--------|-------------|-----------|------------|----------------------------------------|
| 1100     | None   | 10          | 3.75      | 3.5        | 11                                     |
| 1100     | H12    | 10          | 3.75      | 11         | 14                                     |
| 2014     | None   | 10.8        | 4         | 8          | 22                                     |
| 2014     | T6     | 10.6        | 4         | 58         | 66                                     |
| 2014     | T62    | 10.8        | 4         | 59         | 67                                     |
| 2017     | T4     | 10.5        | 3.95      | 32         | 55                                     |
| 2024     | T3     | 10.6        | 4         | 42         | 64                                     |
| 2024     | T4     | 10.6        | 4         | 40         | 62                                     |
| 2025     | T6     | 10.4        | 3.9       | 33         | 52                                     |
| 2124     | T851   | 10.4        | 4         | 57         | 66                                     |
| 2219     | T62    | 10.5        | 4         | 36         | 54                                     |
| 2618     | T61    | 10.7        | 4.1       | 42         | 55                                     |
| 3003     | None   | 10          | 3.75      | 5          | 14                                     |
| 3003     | H18    | 10          | 3.75      | 25         | 27                                     |
| 3003     | H112   | 10          | 3.75      | 10         | 17                                     |
| 354      | T61    | 10.6        | 4         | 36         | 47                                     |
| 355      | T51    | 10.2        | 3.8       | 22         | 27                                     |
| 355      | T6     | 10.3        | 3.8       | 23         | 37                                     |
| 356      | T7     | 10.3        | 3.85      | 21         | 29                                     |
| 356      | T6     | 10.3        | 3.85      | 20         | 30                                     |
| 5052     | None   | 10.2        | 3.8       | 9.5        | 25                                     |
| 5052     | H32    | 10.2        | 3.8       | 23         | 31                                     |
| 5056     | H38    | 10.3        | 3.75      | 50         | 60                                     |
| 5056     | ANL    | 10.3        | 4         | 22         | 42                                     |
| 6061     | T4     | 10          | 3.8       | 16         | 30                                     |
| 6061     | T6     | 10          | 3.8       | 35         | 42                                     |
| 6062     | None   | 10          | 3.75      | 5          | 14                                     |
| 6062     | T4     | 10          | 3.75      | 16         | 26                                     |
| 6062     | T6     | 10          | 3.75      | 35         | 38                                     |
| 6063     | T42    | 10          | 3.75      | 10         | 17                                     |
| 6063     | T5     | 10          | 3.75      | 16         | 22                                     |
| 6063     | T6     | 10          | 3.75      | 25         | 30                                     |
| 6151     | T6     | 10.1        | 3.85      | 37         | 44                                     |
| 7050     | T7452  | 10.1 - 11.6 | None      | 58         | 68                                     |
| 7050     | T74    | None        | None      | 60         | 70                                     |
| 7075     | T6     | 10.4        | 3.9       | 70         | 78                                     |

This script uses the requests and BeautifulSoup libraries to download and parse a webpage from Engineering ToolBox to extract a specific data table containing aluminum pipe properties. It iterates through the HTML table rows to clean the scraped text, replacing empty cells with placeholders and converting numerical strings into mathematical float values. Finally, the program uses the openpyxl library to write this organized data into a new Excel workbook and saves the resulting .xlsx spreadsheet directly to the local directory.